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Industrial Hemp Response to Waterlogging Stress: Influence of Duration and Growth Stage on Physiological Performance and Yield

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Abstract

Industrial hemp is often considered sensitive to waterlogging; however, empirical evidence remains scarce. This study evaluated the effects of waterlogging duration and timing on the physiology, growth, and development of hemp under controlled conditions. Two glasshouse experiments were conducted using industrial hemp (*Cannabis sativa* L.) cv. Ferimon 12. The first examined five waterlogging durations: 0 (control), 3, 6, 12, and 24 days, imposed at the eight true leaf pair stage. The second examined 3- and 6-day waterlogging events imposed at the six true leaf pair and flower initiation stages. In Experiment 1, waterlogging beyond 6 days caused severe damage, and plants died after 12 days of continuous waterlogging. Plants subjected to 3–6 days of waterlogging survived, but photosynthetic activity declined by 40–80%, biomass by 18–55%, and seed yield by 16–48% relative to the control, with the greatest reductions occurring after 6 days of waterlogging. Experiment 2 showed that plant sensitivity varied with developmental stage, with plants exhibiting greater susceptibility at the six true leaf pair stage compared to the flower initiation stage. The dual-purpose hemp cultivar Ferimon 12 was highly sensitive to waterlogging, particularly during early vegetative growth, with even short-term saturation impairing photosynthesis, stomatal conductance, growth and yield. These findings characterise the physiological limits of this cultivar under controlled conditions and demonstrate that developmental stage influences the severity of waterlogging impacts.

Keywords: industrial hemp; waterlogging duration; growth stage; physiological response; growth response; yield

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1. Introduction

Industrial hemp (*Cannabis sativa* L.) is one of the world's oldest cultivated crops and has been widely adopted due to its versatility. It is used in textiles, bioplastics, construction materials, nutritional products, and medicinal purposes, with cannabidiol rich varieties valued for their therapeutic properties. The growing demand for hemp-derived products, together with the crop's broad adaptability and the easing of cultivation restrictions in many regions, has renewed interest in understanding how environmental conditions and agronomic management practices influence hemp growth and productivity.

Hemp requires adequate moisture during establishment, particularly within the first six weeks [1]. However, excessive water can reduce growth, depress yield, and even cause crop failure. Waterlogging induces prolonged rootzone saturation and oxygen deficiency,

impairing root respiration, nutrient uptake, and plant development [2]. Consequently, well-drained soils and the avoidance of prolonged saturation are essential for optimal hemp growth. Under anaerobic conditions, plant roots shift from aerobic respiration to less efficient fermentation pathways, resulting in lower energy production [3] and the accumulation of toxic metabolites that damage root cells [3]. Root hypoxia or anoxia further triggers physiological responses such as increased abscisic acid (ABA) levels, leading to stomatal closure, reduced water and nutrient uptake, decreased photosynthetic rates, and ultimately suppressed growth and wilting [4,5]. Waterlogging further decreases chlorophyll content and chlorophyll fluorescence, indicating impaired photosynthetic performance [6,7].

Numerous studies have investigated the effects of water stress (drought) on the physiology, growth, and yield of industrial hemp [8–15]. In contrast, research on waterlogging in hemp remains limited [16], with much of the available evidence being anecdotal and lacking systematic evaluation. Although waterlogging effects have been extensively studied in other crops such as cotton, wheat, and barley [17–21], comparatively few studies have addressed how hemp responds to waterlogging conditions [16].

The impact of waterlogging on crop performance varies with crop sensitivity, the duration of waterlogging and the growth stage at which it occurs [22]. Kumar et al. reported that a short duration of waterlogging at the vegetative stage of industrial hemp in a field trial reduced photosynthetic traits; however, these traits recovered by the end of the growth stage [16]. In contrast, waterlogging during critical reproductive phases has been shown in other crops, such as cotton, to significantly reduce plant height, leaf area, stem diameter, and yield [17,21,23,24]. Wang et al. found that waterlogging effects increase with duration and are highly stage-dependent in cotton, with the flowering stage being the most sensitive, where even 2 days of waterlogging can cause irreversible damage, while plants at early growth stages can recover from short-term stress [21]. The flowering stage is highly sensitive to waterlogging because reproductive development requires substantial energy and nutrient supply, which is compromised by oxygen deficiency, reduced root function, decreased photosynthesis, and hormonal imbalances under stress, ultimately leading to flower abortion and irreversible yield loss [25].

While short term exposure may allow some degree of recovery, prolonged waterlogging generally results in substantial reductions in biomass accumulation and final yield in other crops [17,21,24,26,27]. Beegum et al. reported that waterlogging beyond 4–6 days initiates significant growth reduction, with severe physiological and biomass losses occurring after 8–10 days, and maximum damage observed at 12–14 days in cotton [17]. Milroy and Bange also observed that photosynthesis in cotton declined significantly after 72 h of waterlogging but recovered once soil oxygen levels improved [23]. They also found that plants showed some acclimation, with photosynthesis becoming less sensitive to low oxygen at later growth stages [23].

In a previous study, Kumar et al. showed that excess water causing waterlogging during the vegetative and flower initiation stages reduced photosynthesis by 20–30% and limited hemp growth in field conditions [16]. This field study established the impact of contrasting water regimes on crop performance. However, a precise evaluation of waterlogging effects requires controlled conditions; therefore, the present experiment was conducted in a pot-based controlled system. Containerized cultivation is widely used in both research and commercial production of cannabis, including medicinal cannabis systems [28–30]. Cannabis and hemp are taxonomically the same species, *Cannabis sativa* L., differing mainly in cannabinoid profile and end use applications. Owing to their similar growth physiology and cultivation practices, including containerised production

systems, the present study has direct relevance to both medicinal cannabis and industrial hemp industries.

Accordingly, this study aims to address the existing knowledge gap by systematically evaluating the effects of waterlogging imposed at different growth stages and durations on hemp grown under controlled glasshouse conditions, with a focus on plant survival, photosynthesis, growth, phenology, and yield.

2. Materials and Methods

Two glasshouse experiments were conducted to evaluate the effects of waterlogging on hemp growth and physiological responses. The first experiment assessed plant tolerance to different durations of waterlogging, while the second experiment was further refined based on initial findings to examine plant responses to moderate waterlogging imposed at different growth stages.

Both experiments used a single genotype grown in a uniform substrate to ensure consistency and minimise genetic and edaphic variability. The use of a substrate also allowed precise control of waterlogging duration and timing by reducing physical and chemical heterogeneity typical of field soils. Using a uniform substrate minimised variability associated with soil physical and chemical properties, allowing plant responses to be compared across waterlogging treatments under controlled conditions. This controlled approach aligns with the study's objective of establishing a foundational understanding of the physiological and morphological responses of hemp to waterlogging.

2.1. Experimental Location and Growth Conditions

The experiments were conducted in a glasshouse at the Horticulture Research Centre at the University of Tasmania, Hobart, Australia. The first experiment was conducted from February to June 2019 and the second experiment from August to December 2019.

Daylength was maintained at 13 h 40 min to delay flowering [31], using natural daylight supplemented, when required, with two 400 W mercury vapour lamps providing approximately 40–80 μmol m^{-2} s^{-1} photosynthetically active radiation (PAR). Supplemental lighting duration was adjusted seasonally to maintain the target photoperiod.

Glasshouse temperatures were monitored using a Tiny Tag data logger. Temperatures were generally maintained through ventilation at an average maximum of ~25 °C and minimum of ~15 °C, which are within the optimal range for hemp growth. However, a few higher temperature events (up to ~31 °C) were recorded due to occasional environmental fluctuations; these instances were infrequent.

The monoecious hemp cultivar Ferimon 12 was selected for this study due to its uniform growth habit and widespread use in industrial hemp production, which improves the relevance of the findings. The use of a single variety further minimised genotypic variability and enabled a clearer interpretation of treatment effects, ensuring that observed differences in growth, physiology, and yield were primarily attributable to waterlogging duration and timing rather than genetic differences.

Seeds were sown in 15 L polypropylene bags (24 cm height) filled with a substrate consisting of a 1:1:1 mix of peat moss, perlite, and vermiculite. The peat–perlite–vermiculite substrate was selected because of its uniform physical properties and rapid drainage under non-waterlogged conditions, allowing the onset and removal of rootzone saturation to be precisely controlled during treatment application. A substrate bulk density of 0.13 g cm^{-3} was calculated as the average of the reported bulk densities of peat moss (0.12 g cm^{-3}), perlite (0.08 g cm^{-3}), and vermiculite (0.19 g cm^{-3}) in the 1:1:1 ($v/v/v$) mixture and was used for unit conversions. This estimate falls within the reported dry bulk density range of

0.08–0.14 g cm^{-3} for peat–perlite substrates [32]. The substrate pH was adjusted to 6.0–6.5 by incorporating dolomite at 36.2 g kg^{-1}, providing Ca and Mg while buffering acidity.

Basal nutrients were applied prior to sowing by adding a controlled-release fertiliser (Osmocote, 7.7 g kg^{-1}, N:P:K ratio 15:4:9 plus micronutrients) in combination with a micronutrient blend (Micromax, 8.5 g kg^{-1}, Fe 6% *w*/*w*, Mn 2.5% *w*/*w*, Zn 1% *w*/*w*, Cu 0.5% *w*/*w*, B 0.1% *w*/*w*, Mo 0.05% *w*/*w*), applied according to manufacturer recommendations to meet nutritional requirements.

Four seeds were sown per pot at ~3 cm depth and thinned to one plant per pot after emergence.

2.2. Treatments

2.2.1. Experiment 1

Waterlogging treatments began at 43 Days After Sowing (DAS) (eight true leaf pair vegetative stage). Before treatment, all plants received overhead sprinkler irrigation (19 L h^{-1}) for 8 min per day, applied in three intervals. Following treatment initiation, control plants continued under the same irrigation regime, while waterlogged plants were maintained under continuous inundation for the duration of their respective waterlogging treatments.

A randomised complete block design was used with six replicates and five waterlogging durations: 0 (control), 3, 6, 12, and 24 days. Waterlogging was imposed by placing individual 15 L pots into 50 L containers and maintaining the water level approximately 2.5 cm above the substrate surface throughout the treatment period. This ensured complete saturation of the potting media and restricted gas exchange between the root zone and atmosphere, creating conditions conducive to rootzone hypoxia. Water levels were checked daily and adjusted as required to maintain consistent inundation. Following each waterlogging period, pots were drained for 24 h without irrigation, after which normal irrigation resumed.

2.2.2. Experiment 2

Similar to Experiment 1, before treatment, all plants received overhead sprinkler irrigation (19 L h^{-1}) for 8 min per day, applied in three intervals. Following treatment initiation, control plants continued under the same irrigation regime, while waterlogged plants were maintained under continuous inundation for the duration of their respective waterlogging treatments. Waterlogging was imposed using the same protocol as Experiment 1, with pots placed in 50 L containers and water maintained approximately 2.5 cm above the substrate surface for the duration of each treatment.

A randomised complete block design was used with five replicates and three waterlogging durations: 0 (control), 3, and 6 days. Waterlogging was applied at two growth stages: six true leaf pair vegetative stage (35 DAS, GS 1012) and flower initiation stage (49 DAS, GS 2301). Following each waterlogging period, pots were drained for 24 h without irrigation, after which normal irrigation resumed.

The treatments therefore included: 0-day waterlogging (control), 3-day waterlogging at the six true leaf pair stage (3 VE), 6-day waterlogging at six true leaf pair vegetative stage (6 VE), 3-day waterlogging at flower initiation stage (3 FI), and 6-day waterlogging at flower initiation stage (6 FI).

2.3. Measurements

Volumetric water content of the substrate was recorded weekly at depths of 4.2, 12.5, and 20.8 cm using an Acclima TDR-315 sensor (Acclima, Meridian, US). Values were summed to provide an integrated measure of substrate water content. Additional measurements were taken during waterlogging periods as required to monitor and maintain the target waterlogging conditions.

From 4 DAS (after seedling emergence), plant development was monitored to calculate thermal time to flower initiation (first visible pistils), end of flowering (95% male flowers withered), and seed maturity (90% hard seeds). Thermal time was calculated using daily maximum and minimum temperatures and a base temperature (Tb) of 5.7 °C, adopted from the base temperature reported for leaf appearance in fibre hemp under controlled-environment conditions [33]. Growth stages followed Mediavilla et al. [34].

Chlorophyll content was measured at 11:00 am on the two youngest fully expanded leaves using a SPAD meter (Apogee Instruments, Logan, UT, USA). Chlorophyll fluorescence (F_v/F_m) was measured after 30 min dark adaptation using a phase amplitude modulated (PAM) fluorometer (OS-30 chlorophyll fluorometer, Opti Sciences, Hudson, NH, USA) on the two youngest fully expanded leaves to assess PSII efficiency.

Net CO_2 assimilation rate (Pn), transpiration rate (Tr), and stomatal conductance (Gs) were measured weekly using a Li-Cor 6400XT infrared gas analyser (IRGA; Li-Cor Biosciences, Lincoln, US). Before measurement, plants were exposed to full sunlight for 1 h near midday. Measurements were taken on the two youngest fully expanded leaves. The IRGA chamber maintained leaf temperature at 20 °C, CO_2 concentration at 400 µmol mol^{-1}, CO_2 flow rate at 400 µmol s^{-1} and PAR at 1500 µmol m^{-2} s^{-1}. Photosynthetic water use efficiency (PWUE) was calculated as Pn/Tr [35].

Plant height and stem diameter were measured weekly (for Experiment 1) and at harvest (for Experiment 1 and Experiment 2). Plant height was measured from the substrate level to the growing tip of the plant using a tape measure. Stem diameter was measured using a Vernier calliper at the mid-point between the first node and the cotyledonary node.

Plants were harvested at the end of the seed maturity; dead and senesced leaves were excluded. In Experiment 1, control and 3-day treatments were harvested at 142 DAS, and the 6-day treatment at 138 DAS. In Experiment 2, control and 3 VE treatments were harvested at 146 DAS. Waterlogged plants matured earlier: 6 VE plants were harvested at 139 DAS (with two plants dying before maturity), and 6 FI plants at 133 DAS.

At each harvest, fresh weight and leaf area were recorded. Bark proportion was determined from stem sections. Stem, leaf, and reproductive tissues were separated and oven dried at 60 °C for 48 h to determine dry yield. Seeds were separated, cleaned and weighed, and total seed yield (g $plant^{-1}$) and thousand seed weight (TSW) were calculated for each treatment.

3. Data Analysis

Repeatedly measured variables, including chlorophyll content, chlorophyll fluorescence (F_v/F_m), net CO_2 assimilation rate (Pn), transpiration rate (Tr), stomatal conductance (Gs), plant height and stem diameter were analysed using the PROC MIXED procedure in SAS version 9.4. The model included treatment, sampling time, and the treatment × time interaction as fixed effects. Individual plants were considered the experimental units and specified as subjects in the repeated-measures analysis to account for correlations among observations collected from the same plant over time. Type III F-tests were used to evaluate fixed effects. When significant treatment × time interactions were detected ($p \leq 0.05$), treatment means were compared within sampling dates using Tukey-adjusted multiple comparisons. Plants that died during the experiment contributed data up to the time of mortality, after which observations were treated as missing. PROC MIXED accommodated missing repeated observations without requiring data imputation. Individual plants ($n = 6$ per treatment in Experiment 1; $n = 5$ per treatment in Experiment 2) were considered the experimental units and specified as subjects in the repeated-measures analysis.

Data from destructive measurements were analysed using analysis of variance (ANOVA) to evaluate the effects of waterlogging on plant responses. The sample size

(n = 5–6 replicates per treatment) was selected based on common practice in controlled-environment and glasshouse studies, as well as logistical constraints associated with the experimental setup. While a formal a priori power analysis was not performed, the study detected significant treatment effects across several key response variables.

In Experiment 1, which included only waterlogging duration treatments (0-control, 3, 6, 12, and 24 days), treatment effects were assessed using one way ANOVA.

In Experiment 2, treatment means for all five treatments (0 days (control), 3 VE, 6 VE, 3 FI, and 6 FI) were first compared using one-way ANOVA to evaluate overall treatment differences relative to the 0 days (control). To specifically examine the effects of waterlogging duration and growth stage, a separate two-way ANOVA was conducted using only waterlogging treatments (3 VE, 6 VE, 3 FI, and 6 FI). Waterlogging duration (3 and 6 days) and growth stage (vegetative and flower initiation) were included as fixed factors. The 0 days (control) was excluded from the factorial analysis because it could not be assigned to either a waterlogging duration or a growth-stage treatment level.

Statistical significance was determined using Type III F tests at $p \leq 0.05$. When interactions were significant, treatment means were separated using Tukey's HSD test. For variables with non-significant interactions, main effects were interpreted and means compared using Tukey's HSD test.

4. Results

4.1. Experiment 1

4.1.1. Volumetric Water Content of the Substrate

Total volumetric water content in the control treatment remained stable throughout the experiment. Waterlogging increased substrate water content by approximately 0.0417 L L^{-1} in the 3-day treatment, 0.0625 L L^{-1} in the 6- and 12-day treatments, and 0.083 L L^{-1} in the 24-day treatment. Plants subjected to 12- and 24 days of waterlogging died by 66 DAS (23 days after waterlogging). After waterlogging ceased, water content declined across all treatments. Differences in substrate water content among treatments were minimal both before and after the waterlogging events. (Figure S1).

4.1.2. Leaf Chlorophyll, CO_2 Exchange, Stomatal Conductance and PWUE

Relative chlorophyll content, Pn, Tr, Gs and PWUE exhibited significant treatment × time interactions ($p < 0.05$), indicating that responses to waterlogging differed among treatments over time, particularly following treatment imposition and during the recovery phase. Relative chlorophyll content, Pn, Tr, Gs and PWUE showed contrasting temporal patterns among waterlogging treatments (Figure 1; Figure S2a,b). Treatment differences were small prior to waterlogging but became apparent shortly after treatment imposition (approximately 53 DAS) and increased during the subsequent recovery period. Control plants maintained the highest values throughout the experiment, whereas plants subjected to 3 and 6 days of waterlogging showed varying rates of recovery following drainage.

Plants exposed to 3 days of waterlogging showed an initial decline in measured variables but recovered rapidly after drainage, subsequently following a pattern broadly similar to the control. In contrast, plants exposed to 6 days of waterlogging exhibited a more prolonged reduction in relative chlorophyll content, Pn, Tr and Gs, with recovery becoming evident only from approximately 74 DAS onwards. These variables gradually increased thereafter and, by the end of the crop cycle, approached values observed in the control and 3-day treatments. However, PWUE remained lower than the control and 3-day treatments for most of the remainder of the crop cycle (Figure 1c).

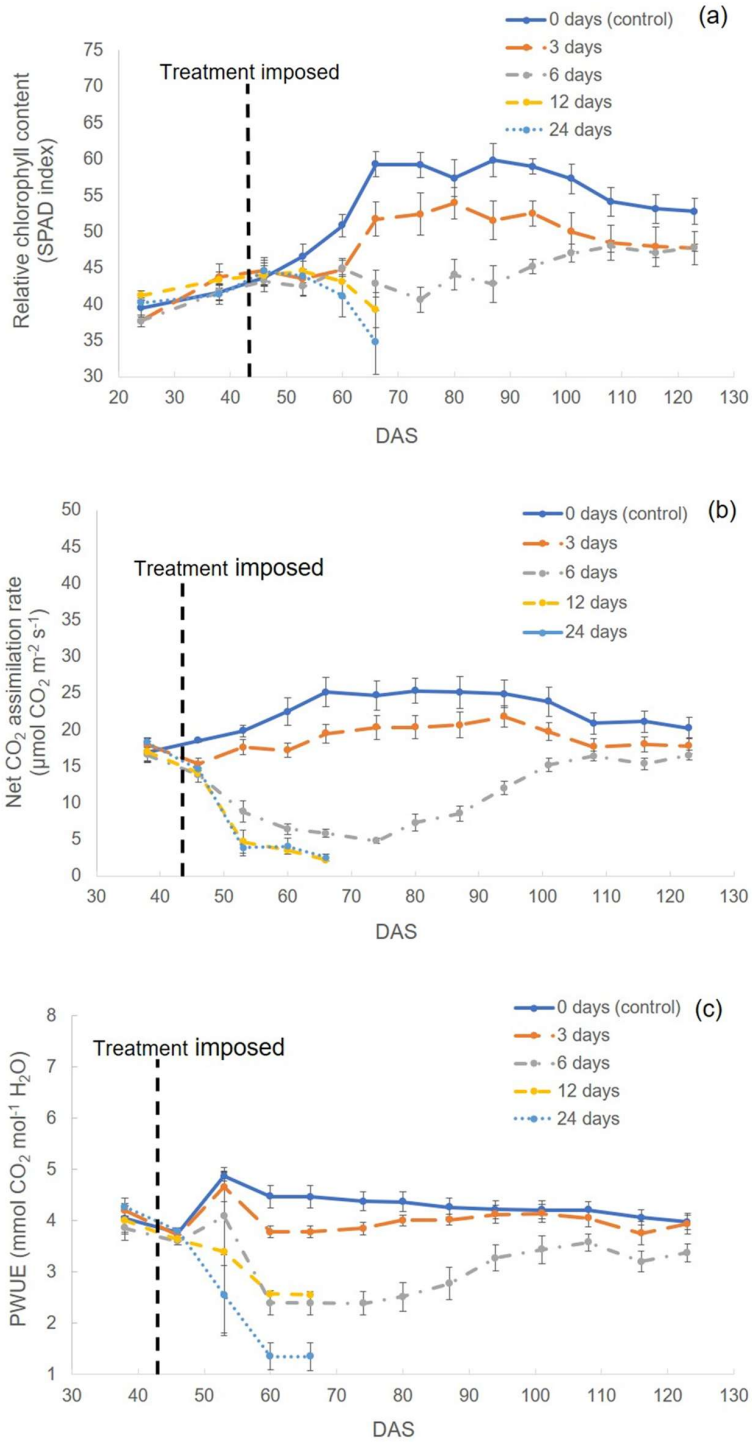


Figure 1. Relative chlorophyll content (SPAD index, (**a**)), Net CO_2 assimilation rate (A_{sat}, (**b**)), and photosynthetic water use efficiency (PWUE, (**c**)) of hemp leaves in Experiment 1. Each value represents the mean ± S.E. (n = 6). Solid blue lines represent the 0-day waterlogging or control treatment; orange long-dash lines represent 3-day waterlogging; grey dash-dot lines represent 6-day waterlogging; yellow dash lines represent 12-day waterlogging; and light blue dotted lines represent 24-day waterlogging.

Plants exposed to 12 and 24 days of waterlogging exhibited rapid declines in relative chlorophyll content, Pn, Tr, Gs and PWUE following treatment imposition. These plants failed to recover after drainage, and mortality occurred approximately 23 days after waterlogging commenced (Figure 1; Figure S2a,b).

4.1.3. Crop Phenology

Waterlogging duration did not affect the calendar and thermal time required for flower initiation. The average thermal time for flower initiation was 662.2 °Cd. However, the duration from flower initiation to the end of flowering was significantly shorter ($p < 0.05$) in the 6-day waterlogging treatment (343 °Cd) compared with the control and 3-day treatments. All plants in the 12- and 24-day treatments died shortly after flowering, whereas plants in the other treatments continued to grow. The thermal time from the end of flowering to seed maturity was consistent across the control, 3-day, and 6-day treatments, with no significant differences (Table S1).

4.1.4. Plant Height and Stem Diameter

In the control, plant height and stem diameter increased steadily until flowering (~66 DAS) and then plateaued. The 3-day waterlogging treatment caused a temporary slowdown of growth, with full recovery by ~94 DAS. In contrast, plants waterlogged for 6 days experienced a prolonged growth setback and only partial recovery, resulting in reduced final height and diameter compared with both the control and 3-day treatments. Plants subjected to 12- and 24-day waterlogging showed irreversible growth suppression and died by 66 DAS (Figure 2).

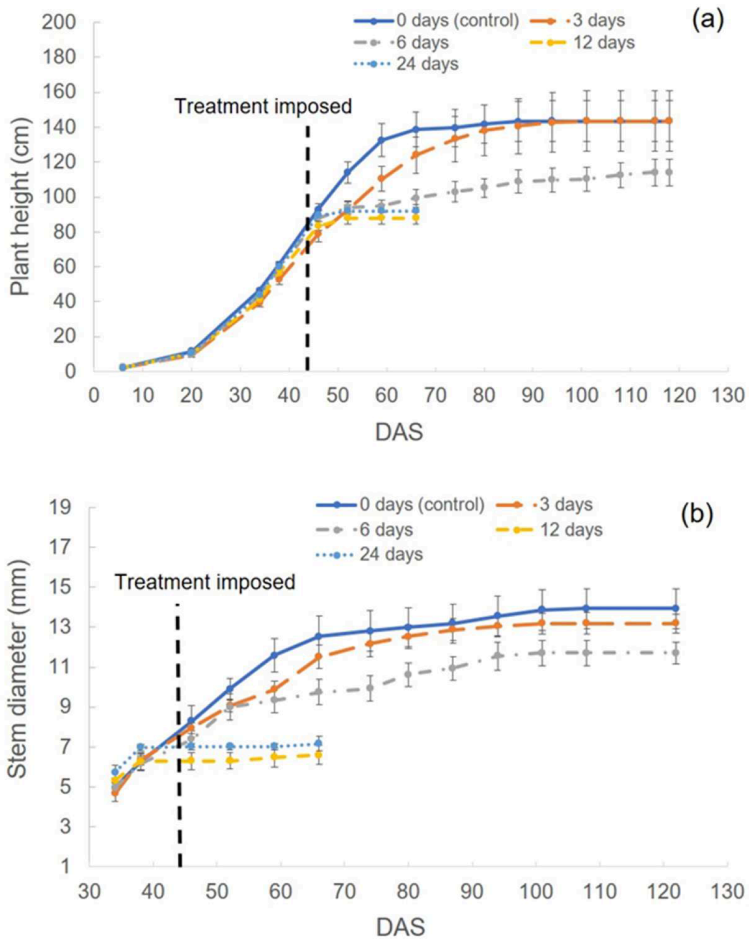


Figure 2. Plant height (**a**) and stem diameter (**b**) in Experiment 1. Each value represents the mean ± S.E. (n = 6). Solid blue lines represent the 0-day waterlogging or control treatment; orange long-dash lines represent 3-day waterlogging; grey dash-dot lines represent 6-day waterlogging; yellow dash lines represent 12-day waterlogging; and light blue dotted lines represent 24-day waterlogging.

4.1.5. Specific Leaf Area (SLA), Dry Biomass, Stalk Yield and Seed Yield

Complete mortality occurred in the 12-day and 24-day waterlogging treatments following treatment imposition, with no plants surviving to final harvest. The control, 3-day, and 6-day waterlogging treatments survived to seed maturity and were included in harvest assessments.

At harvest, SLA declined significantly as waterlogging duration increased from 0 to 6 days ($p < 0.05$). Similarly, leaf, stem, and total aboveground dry weights (TADW) decreased with longer waterlogging, with the 6-day treatment producing significantly lower values than both the control and 3-day treatments ($p < 0.05$) (Table 1).

Table 1. Mean SLA, dry biomass production, stalk yield and seed yield at harvest in Experiment 1.

Waterlogging Duration (Days)	SLA (m^2 kg^{-1})	DW Leaves (g $Plant^{-1}$)	DW Stems (g $Plant^{-1}$)	TADW (g $Plant^{-1}$)	Stalk Yield (g $Plant^{-1}$)	Seed Yield (g $Plant^{-1}$)
0 (control)	15.0 [a]	21.1 [a]	40.1 [a]	77.0 [a]	15.0 [a]	8.2 [a]
3	13.7 [b]	16.3 [a]	34.4 [a]	62.4 [a]	11.0 [a]	6.9 [a]
6	8.6 [c]	10.0 [b]	16.8 [b]	34.7 [b]	5.0 [b]	4.3 [b]

Mean value of data. Within a column, data means followed by the different letters are significantly different at the $p < 0.05$ level. SLA, specific leaf area; DW, dry weight; TADW, total above ground dry weight.

Waterlogging had no significant effect on bark proportion at seed maturity. However, both bark and seed yield declined with waterlogging duration, with the lowest yields recorded in the 6-day treatment ($p < 0.05$) (Table 1). These results indicate that plants exposed to 3 days of waterlogging were able to largely recover from the imposed stress, with final growth and yield comparable to the control. In contrast, although plants exposed to 6 days of waterlogging showed partial recovery of physiological function following drainage, reductions in plant size, biomass accumulation, and yield persisted until harvest.

4.2. Experiment 2

4.2.1. Volumetric Water Content of the Substrate

Water content in the control treatment (0-day) remained relatively consistent throughout the experiment. In the waterlogging treatments, substrate water content increased by ~0.083 L L^{-1} following waterlogging and then declined after the waterlogging conditions were removed. Similar to Experiment 1, differences in water content among treatments were minimal both before and after the waterlogging events (Figure S3).

4.2.2. Leaf Chlorophyll, CO_2 Exchange, Stomatal Conductance and PWUE

Relative chlorophyll content, F_v/F_m, net CO_2 assimilation rate (Pn), transpiration rate (Tr), stomatal conductance (Gs), and PWUE were all significantly influenced by the treatment × time interaction ($p < 0.05$), indicating that the temporal response differed among waterlogging treatments. Following the initial decline during waterlogging, plants subjected to 3-day waterlogging (3 VE and 3 FI) recovered rapidly and subsequently followed trajectories similar to the control for relative chlorophyll content, F_v/F_m, Pn, Tr and Gs (Figure 3a–c; Figure S4a,b). In contrast, plants exposed to 6 days of waterlogging (6 VE and 6 FI) exhibited more pronounced reductions and slower recovery, resulting in significant differences from the control during the early post-waterlogging period before converging later in the season (Figure 3a–c; Figure S4a,b).

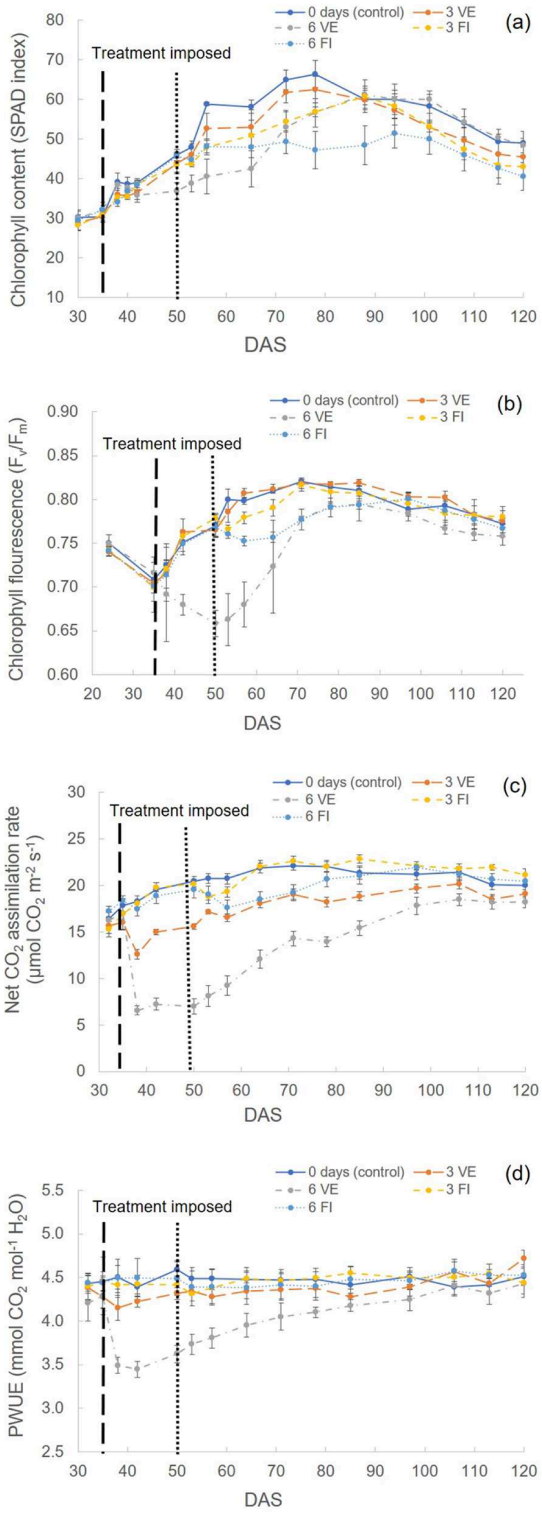


Figure 3. Relative chlorophyll content (SPAD index, (**a**)), chlorophyll fluorescence (**b**), PAR-saturated net CO_2 assimilation rate (A_{sat}, (**c**)) and photosynthetic water use efficiency (PWUE, (**d**)) of hemp leaves in Experiment 2. Each value is the mean ± S.E. (n = 5). The dashed lines at 35 DAS and 49 DAS indicate the initiation of waterlogging (WL) events at the six true leaf pair vegetative stage and flower initiation stage, respectively, for the 3-day and 6-day treatments. Solid blue lines represent the 0-day waterlogging or control treatment; orange long-dash lines represent 3-day waterlogging at six true leaf pair vegetative stage (3 VE); grey dash-dot lines represent 6-day waterlogging at six true leaf pair vegetative stage (6 VE); yellow dash lines represent 3-day waterlogging at flower initiation stage (3 FI); and light blue dotted lines represent 6-day waterlogging at flower initiation stage (6 FI).

PWUE in plants waterlogged at flower initiation (3 FI and 6 FI) was comparable to the control throughout the experiment. The 3 VE plants showed slightly lower PWUE until 105 DAS, while 6 VE plants exhibited a marked reduction immediately after waterlogging, followed by gradual recovery after 40 DAS and convergence with the control by 105 DAS (Figure 3d).

4.2.3. Crop Phenology

Flower initiation occurred 7 days earlier ($p < 0.05$) in 6 VE plants, which required 517 °Cd of thermal time compared with the other treatments. The 6 FI plants progressed more rapidly ($p < 0.05$) through both subsequent developmental phases, requiring ~335 °Cd from flower initiation to the end of flowering and ~847 °Cd from the end of flowering to seed maturity (Table S2).

4.2.4. Specific Leaf Area (SLA), Dry Biomass, Plant Height, Stem Diameter, Stalk Yield and Seed Yield

Compared with the 0 days (control), all waterlogging treatments reduced one or more growth and yield parameters at final harvest (Table 2). The greatest reductions occurred in the 6 VE treatment, where SLA, total above-ground dry weight, stalk yield, and seed yield were reduced by approximately 36%, 68%, 83%, and 63%, respectively, relative to the control. Plant height and stem diameter were also reduced by 58% and 53%, respectively. In contrast, plants exposed to waterlogging at flower initiation generally exhibited smaller reductions than those waterlogged during the six true leaf pair vegetative stage. For example, total above-ground dry weight and seed yield in the 3 FI treatment were reduced by only 32% and 13%, respectively, and seed yield did not differ significantly from the control.

Table 2. Growth, dry biomass, stalk yield, and seed yield responses to waterlogging treatments at final harvest in Experiment 2.

Treatments	SLA (m^2 kg^{-1})	Dry Weight of Leaves (g $Plant^{-1}$)	Dry Weight of Stems (g $Plant^{-1}$)	Total Above Ground Dry Weight (g $Plant^{-1}$)	Plant Height (cm)	Stem Diameter (mm)	Stalk Yield (g $Plant^{-1}$)	Seed Yield (g $Plant^{-1}$)
0 Days (Control)	14 ^{a}	25 ^{a}	46 ^{a}	91 ^{a}	151 ^{a}	15 ^{a}	18 ^{a}	8 ^{a}
3 VE	14 ^{a}	20 ^{ab}	23 ^{b}	58 ^{b}	116 ^{b}	12 ^{b}	11 ^{b}	6 ^{b}
6 VE	9 ^{c}	7 ^{d}	7 ^{d}	29 ^{d}	64 ^{c}	7 ^{d}	3 ^{d}	3 ^{d}
3 FI	13 ^{ab}	18 ^{b}	28 ^{b}	62 ^{b}	129 ^{b}	14 ^{ab}	12 ^{b}	7 ^{ab}
6 FI	13 ^{b}	15 ^{c}	18 ^{c}	42 ^{c}	118 ^{b}	11 ^{c}	6 ^{c}	5 ^{c}

Values are means ($n = 5$). Means followed by different lowercase letters within a column are significantly different at $p < 0.05$ based on one-way ANOVA across all treatments followed by Tukey's HSD multiple comparison test.

An overview of the statistical analysis indicated that waterlogging duration and growth stage significantly influenced most measured traits. Interaction effects between duration and growth stage were limited, with significance detected only for SLA (Table 3). Detailed responses of individual traits are described in the following sections.

SLA declined most strongly in the 6 VE treatment, whereas SLA at flower initiation remained relatively stable even after prolonged waterlogging (Table 4).

For all other variables, only main effects were significant (Table 5). Increasing waterlogging duration from 3 to 6 days reduced leaf dry weight, stem dry weight, TADW, plant height, stem diameter, stalk yield, and seed yield, while plants waterlogged at flower initiation consistently showed higher stem DW, TADW, plant height, stem diameter and seed yield than those waterlogged during the six true leaf pair vegetative stage (Table 5).

Table 3. Two-way ANOVA interaction-test for the measured traits in Experiment 2.

	Waterlogging Duration (D)	Growth Stage (S)	D × S
SLA	***	***	***
DW leaves	**	ns	ns
DW stems	**	*	ns
TADW	***	*	ns
Height	***	**	ns
Diameter	**	*	ns
Stalk yield	***	ns	ns
Seed yield	**	*	ns

ns = not significant; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. SLA, specific leaf area; DW, dry weight; TADW, total above ground dry weight.

Table 4. Interaction of waterlogging duration and growth stage on SLA in Experiment 2.

Treatment	SLA (m^2 kg^{-1})
3 VE	14.00 [a]
6 VE	8.70 [c]
3 FI	13.15 [ab]
6 FI	13.08 [b]

Means followed by different lowercase letters indicate significant differences among waterlogging duration × growth stage combinations at $p \leq 0.05$ (two-way ANOVA; interaction effect significant, followed by post hoc multiple comparison test). SLA, specific leaf area; 3 VE, 3-day waterlogging at six true leaf pair; 6 VE, 6-day waterlogging at six true leaf pair; 3 FI, 3-day waterlogging at flower initiation, 6 FI, 6-day waterlogging at flower initiation.

Table 5. Main effect of waterlogging duration and growth stage on measured traits in Experiment 2.

	Waterlogging Duration	
	3 Days	6 Days
DW leaves (g)	19.52 [a]	11.22 [b]
DW stems (g)	27.39 [a]	13.01 [b]
TADW (g)	61.86 [a]	38.76 [b]
Height (cm)	125.60 [a]	100.40 [b]
Diameter (mm)	12.74 [a]	10.01 [b]
Stalk yield (g)	11.57 [a]	5.10 [b]
Seed yield (g)	6.46 [a]	4.02 [b]
	Growth stage	
	VE	FI
DW stems (g)	12.41 [b]	28.00 [a]
TADW (g)	44.23 [b]	56.39 [a]
Height (cm)	102.40 [b]	123.60 [a]
Diameter (mm)	9.43 [b]	12.92 [a]
Seed yield (g)	4.40 [b]	6.08 [a]

Values are means. Within each row, means followed by different lowercase letters differ significantly at $p \leq 0.05$ based on ANOVA and Tukey's HSD test. SLA, specific leaf area; DW, dry weight; TADW, total above ground dry weight. VE, six true leaf pair at the vegetative stage; FI, flower initiation stage.

5. Discussion

In these experiments, hemp, like many other crops, showed poor tolerance to prolonged waterlogging, particularly when saturation occurred early in development. Plant growth and development declined progressively as waterlogging duration increased, regardless of growth stage at which it was imposed. Plants subjected to more than 12 days of waterlogging at the eight true leaf pair stage did not survive, confirming severe sensitivity to extended rootzone saturation. In contrast, plants exposed to shorter periods of waterlogging (3–6 days) during the vegetative phase recovered, although recovery was slower after

6 days. Despite this recovery, long-term reductions in photosynthesis, growth, dry matter production, and yield were observed compared with the control.

The physiological impairment underlying these responses is consistent with well-established effects of hypoxia on root function. Waterlogging restricts oxygen diffusion and creates hypoxic or anoxic conditions that inhibit root respiration, reduce energy production, and impair nutrient uptake. Prolonged oxygen deprivation also causes oxidative damage to root tissues [3]. Although redox potential (Eh) was not measured in the present study, complete inundation of the peat-based substrate would have progressively lowered substrate Eh, creating increasingly reducing conditions. In peat systems, Eh values under inundated conditions have been reported to be approximately 150–200 mV and may decline to below 50 mV under strongly reducing conditions [36]. Such low Eh values are associated with restricted root respiration, altered nutrient dynamics, and reduced plant growth, providing a quantitative indication of the oxygen-deprived conditions likely experienced by hemp in this study. These below ground stresses disrupt canopy development and reduce photosynthesis and yield, as reported in other crops exposed to waterlogging [4–7].

Given the limited comprehensive research on waterlogging tolerance in hemp, comparisons with other crops provide useful context. Tolerance thresholds differ widely among species; wheat and barley can survive and recover from 14 to 16 days of waterlogging without complete plant loss [37,38], whereas durum wheat shows yield reductions only beyond 20 days of saturation [20]. Cotton, by comparison, exhibits far lower tolerance; Zhang et al. reported complete mortality in some genotypes after more than 10 days of waterlogging [39]. In the present study, hemp yield declined by more than 50% after only 6 days of waterlogging, indicating a level of sensitivity comparable to or exceeding that of cotton and considerably lower than that of most cereals.

Physiologically, waterlogging in hemp caused marked reductions in chlorophyll content, Pn, Tr, Gs, and F_v/F_m, consistent with responses reported for cotton, barley, and wheat [18,19,21,27,39–41]. Ren et al. have shown that 6 days of waterlogging at the 3-leaf stage in maize reduced leaf area index, chlorophyll content and photosynthesis, with responses similar to those observed in hemp after 6 days [42]. Mahmood et al. reported that 7 days of waterlogging in maize reduced chlorophyll content to a degree comparable to the reduction observed here in hemp after 6 days [43].

Reductions in Tr and Gs were also notable. In hemp, a 15–25% decline occurred after only 3 days of waterlogging during the most sensitive developmental stage (six true leaf pair). Comparable responses have been reported in other crops; Kubota et al. (2024) observed reduced Tr and Gs in soybean after more than 3 days of waterlogging [44]. Consistent responses were observed by Ploschuk et al. (2018), who observed declines in photosynthetic parameters in barley and rapeseed following 14 days of waterlogging at late growth stages, and in field pea at both early and late developmental stages [40].

Reductions in photosynthetic performance translated directly into losses in biomass accumulation, TADW, and yield. Although yield declines in hemp were greater than those commonly observed in cereals, the overall pattern aligns with findings across species: as waterlogging duration increases, growth and productivity decline proportionally. For example, Pampana et al. (2016) showed a 19–30% reduction in yield in durum wheat after 40–60 days of waterlogging, whereas hemp in this study exhibited >50% yield loss after only 6 days of waterlogging, further highlighting its sensitivity [20].

SLA in hemp declined progressively with increasing waterlogging duration. The greatest reduction (~40%) occurred after 6 days of waterlogging at the eight true leaf pair vegetative stage in Experiment 1, similar to reductions reported in wheat, barley, and maize under comparable stress [20,45].

In hemp, the greater sensitivity to waterlogging at the six true leaf pair vegetative stage compared with the flower initiation stage can be attributed to the importance of early vegetative growth for plant establishment. During this stage, rapid canopy expansion and root development create a high demand for oxygen and nutrient uptake. Consequently, prolonged rootzone saturation may reduce oxygen availability, impair root function, restrict photosynthesis, and ultimately limit plant growth. Although rootzone oxygen status was not measured directly in the present study, the observed responses are consistent with those reported for plants exposed to hypoxic conditions [39,46–48]. In contrast, plants at flower initiation generally have a more established root system and have already accumulated a substantial proportion of their vegetative biomass, allowing greater resilience to short-term stress [39,48]. This pattern differs from crops such as cotton, where waterlogging during flowering causes the greatest yield loss due to its direct impact on reproductive processes and boll development [21,24]. The contrasting responses reflect crop specific growth strategies, as hemp yield particularly in fibre types is largely determined by vegetative biomass accumulation, making early-stage stress more critical than stress imposed during later reproductive development.

Waterlogging also accelerated hemp phenology, with earlier flower initiation and faster progression through reproductive stages in severely stressed plants. These findings are consistent with Zhang et al. (2025) [39], who reported that late stage waterlogging hastened maturity in rapeseed but differs from other studies showing delayed development in wheat and cotton. For example, Wang et al. (2017) found that 6 days of waterlogging at the seedling stage did not affect cotton phenology [21], while Celedonio et al. (2016) reported that waterlogging delayed flowering and reduced tiller appearance in wheat cultivars [18]. Such variation underscores species-specific and stage-specific responses to waterlogging and suggests that hemp may shift phenology as a stress-avoidance mechanism under conditions of prolonged rootzone saturation.

Overall, this study highlights the pronounced susceptibility of hemp to waterlogging stress and demonstrates that both the duration and developmental timing of saturation strongly influence physiological function, growth, and yield.

Although these findings are not intended to directly represent field behaviour, they provide a physiological and stage-specific understanding of hemp responses to transient rootzone saturation. This information is relevant to controlled environment and nursery production systems commonly used for hemp and closely related crops, such as medicinal cannabis, where irrigation management is critical and preventing substrate saturation can help minimise growth and yield penalties.

6. Future Research Directions

The findings of this study provide a foundation for further investigation into hemp responses to waterlogging. Future research could incorporate measurements of dissolved oxygen and substrate redox potential, alongside physiological and growth assessments, to improve characterisation of the rootzone environment and enhance understanding of tolerance and recovery mechanisms under waterlogging conditions.

The results reported here are based on a single hemp cultivar (Ferimon 12) and therefore may not fully represent the diversity of responses within *Cannabis sativa*. The high sensitivity observed in this cultivar may reflect a cultivar-specific response rather than the full spectrum of waterlogging tolerance available within the species. Future research should evaluate waterlogging tolerance across a broader range of fibre, grain, dual-purpose, and dioecious cultivars, as well as southern-adapted landraces, to characterise genotypic variation in physiological responses, recovery potential, and overall resilience to waterlogging stress. Such investigations would contribute to the identification of waterlogging-tolerant

germplasm and support cultivar selection and breeding for environments prone to transient flooding events.

The present study focused on the physiological responses of hemp to waterlogging stress and did not include anatomical assessments. Future studies should investigate anatomical adaptations associated with waterlogging tolerance, including the formation of hypertrophied lenticels, aerenchyma, and adventitious roots, as these traits may contribute to recovery following short term waterlogging. Integrating anatomical and physiological assessments would provide a more comprehensive understanding of the mechanisms underlying waterlogging tolerance and adaptation in hemp.

Future research is also needed to assess fibre quality characteristics, including tensile strength, fibre composition, and cellulose content. Such analyses would provide a more comprehensive understanding of the impacts of waterlogging on hemp fibre production and quality.

7. Conclusions

This study demonstrates that the dual-purpose hemp cultivar Ferimon 12 is highly susceptible to waterlogging, with both the duration of saturation and the developmental stage at which it occurs exerting strong effects on physiological function, growth, and yield. Even short periods of waterlogging during the early vegetative stage (six true leaf pair) caused lasting reductions in photosynthesis, biomass accumulation, and seed yield, while prolonged waterlogging resulted in complete plant mortality. Compared with many cereals, hemp displayed markedly lower tolerance to waterlogging, with 6 days of waterlogging reducing total aboveground biomass by more than 50% and seed yield by approximately 48%.

To optimise fibre and seed yield, prolonged waterlogging and the associated rootzone saturation should be minimised, particularly during early vegetative growth. Successful hemp cultivation requires well-aerated, free-draining soils or growing media and careful water management to minimise prolonged soil saturation and the associated risk of reduced oxygen availability in the rootzone. Although this study was conducted using a single substrate under controlled glasshouse conditions, the results clearly demonstrate the importance of waterlogging duration and timing in determining hemp performance.

This work provides a physiological, stage-specific understanding of hemp responses to waterlogging stress and offers a foundation for improving management strategies in both field and controlled-environment production. It also highlights the need for future breeding and agronomic research aimed at enhancing waterlogging tolerance in hemp.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/crops6050090/s1, Figure S1: Total volumetric soil moisture content (L L^{-1}) from 0 cm to 24 cm depth during crop growth in Experiment 1; Figure S2: Transpiration rate (a) and stomatal conductance (b) of hemp leaves in Experiment 1; Table S1: Calendar (days) and thermal time durations for key crop growth developmental phases for the different treatments in Experiment 1; Figure S3: Total volumetric soil moisture content (L L^{-1}) from 0 cm to 24 cm depth during crop growth in Experiment 2; Figure S4: Transpiration rate (a) and stomatal conductance (b) of hemp leaves in Experiment 2; Table S2: Calendar (days) and thermal time durations for key crop growth developmental phases for the different treatments in Experiment 2.

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Abbreviations

The following abbreviations are used in this manuscript:

ANOVA	Analysis of variance
DAS	Days after sowing
DW	Dry weight
FI	Flower Initiation
F_v/F_m	Chlorophyll fluorescence
Gs	Stomatal conductance
IRGA	Infrared gas analyser
PAM	Phase amplitude modulated
PAR	Photosynthetically active radiation
Pn	Net CO_2 assimilation rate
PWUE	Photosynthetic water use efficiency
SE	Standard error
SLA	Specific leaf area
TADW	Total above ground dry weight
Tr	Transpiration rate
TSW	Thousand seed weight
VE	Vegetative stage

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